

Advanced (Habanero)

- Q1.** Solve the system: $x^2 + y^2 = 4$, $x^2 - y^2 = -1$
(A) $(-1, -\sqrt{3})$
(B) $(-1, \sqrt{3})$
(C) $(1, -\sqrt{3})$
(D) $(1, \sqrt{3})$
(E) No solution
- Q2.** What is the solution to the system: $y = x^2 - 4$, $y = -x^2 + 4$?
(A) $(0, 0)$
(B) $(0, 4)$
(C) $(2, 0)$
(D) $(-2, 0)$
(E) Infinite solutions
- Q3.** Solve the system: $x^2 + y^2 = 25$, $y = x^2 - 5$
(A) $(-2, 1)$
(B) $(-3, 4)$
(C) $(2, 1)$
(D) $(3, 4)$
(E) No real solutions
- Q4.** The graphs of $y = x^2 - 2x - 3$ and $y = x^2 + 2x - 3$ intersect at what points?
(A) $(-1, -4)$, $(1, -4)$
(B) $(-1, 0)$, $(1, 0)$
(C) $(-3, 0)$, $(3, 0)$
(D) $(-2, -1)$, $(2, -1)$
(E) No real solutions
- Q5.** What is the number of solutions for the system: $x^2 + y^2 =$ 16, $x^2 + y^2 = 25$?
(A) 0
(B) 1
(C) 2
(D) 4
(E) Infinite
- Q6.** The system $x^2 - y^2 = 4$, $x^2 + y^2 = 8$ has how many solutions?
(A) 0
(B) 1
(C) 2
(D) 4
(E) Infinite
- Q7.** The graphs of $y = x^2 - 4x + 3$ and $y = -x^2 + 4x - 3$ intersect at what x -value?
(A) 0
(B) 1
(C) 2
(D) 3
(E) 4
- Q8.** The system of equations $x^2 + 2y^2 = 6$, $2x^2 + y^2 = 6$ has what kind of solution set?
(A) 1 solution
(B) 2 solutions
(C) Infinite solutions
(D) No real solutions
(E) Dependent

Open Ended Questions (FRQ)

- Q9.** Solve the system $y = x^2 - 3x - 2$, $y = -x^2 - 3x + 2$ and find the intersection points. Show all work.
- Q10.** Graph the system $x^2 + y^2 = 9$, $y = x^2 - 4$. Label all intersection points and axes.

Chef's Tips

- Check for correct units and labels
- Ensure all work is shown for FRQs
- Use a ruler for graphing